Convolutional Neural Networks for CIFAR-10 Image Recognition

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Abstract

This report presents the design, training, and analysis of a Convolutional Neural Network (CNN) on the CIFAR-10 dataset. We explore preprocessing and augmentation strategies, investigate how filter size and stride influence performance, and study the impact of L2 regularization on generalization. The analysis includes learning curves, weight and bias distributions, correct/misclassified samples, and intermediate feature-map visualizations.

1 Dataset and Preprocessing

CIFAR-10 consists of 60,000 RGB images (32×32×3) in ten classes: airplane, automobile, bird, cat, deer, dog, frog, horse, ship, and truck. There are 50,000 training and 10,000 testing images.

1.1 Data Normalization

Each pixel was scaled to the range [0,1] by dividing by 255. This ensures that all channels have a comparable scale, improving convergence speed and numerical stability.

1.2 Data Splitting

From the 50,000 training images, 5,000 were reserved as a validation set for hyperparameter tuning and early stopping:

$$(x_{\text{val}}, y_{\text{val}}) = (x_{\text{train}}[: 5000], y_{\text{train}}[: 5000])$$

and the remaining 45,000 images were used for training.

1.3 Data Augmentation

To reduce overfitting and improve generalization, `ImageDataGenerator` was used with:

- Random horizontal flips.
- Random rotations up to ±15°.
- Width and height shifts of up to 10%.
- Random zoom within 10%.

Augmentation was applied only to training images.

2 Model Architecture

2.1 Network Design

We used a compact CNN architecture optimized for CIFAR-10 resolution:

- Two convolutional blocks:
  - Each block contains two `Conv2D` layers followed by `BatchNormalization`, `MaxPooling2D`, and `Dropout`.
- Dense classifier:
  - Flatten layer, fully connected layer (128 units, ReLU), dropout, and final softmax layer.

Default parameters: kernel sizes (3,3), stride (1,1), dropout 0.4, He-normal initialization, and Adam optimizer.

2.2 Regularized Model

An L2-regularized variant adds $\lambda\|W\|_2^2$ to the loss for all `Conv2D` and `Dense` layers. This discourages large weight magnitudes, helping reduce overfitting.

3 Experimental Setup

- Optimizer: Adam (default learning rate 10^{-3})
- Loss: Sparse categorical cross-entropy
- Epochs: 15
- Batch size: 128
- Metrics: Accuracy and cross-entropy loss

4 Results and Discussion

4.1 Quantitative Comparison

Table 1 summarizes the performance for various configurations.

Table 1: Effect of filter size, stride, and L2 regularization on CIFAR-10 performance (15 epochs).

Model	**Kernel**	**Stride**	**Train Acc**	**Val Acc**	**Train Loss**	**Val Loss**
Baseline	(3,3)	(1,1)	0.7846	**0.7908**	0.6223	0.6352
Large Stride	(5,5)	(2,2)	0.7422	0.7486	0.7331	0.7128
Large Kernel	(7,7)	(1,1)	**0.8189**	0.7916	**0.5285**	0.6162
L2 Regularized	(7,7)	(1,1)	0.7339	0.7212	1.1238	1.1836

Analysis. Smaller kernels (3×3) balance detail capture and efficiency, achieving strong validation accuracy. Large kernels (7×7) enlarge the receptive field and improve training accuracy but only slightly improve validation performance. Larger strides (2,2) aggressively downsample and lose spatial detail, reducing accuracy. Adding L2 regularization smooths the weight distributions and reduces overfitting but can underfit if λ is too high.

4.2 Learning Curves

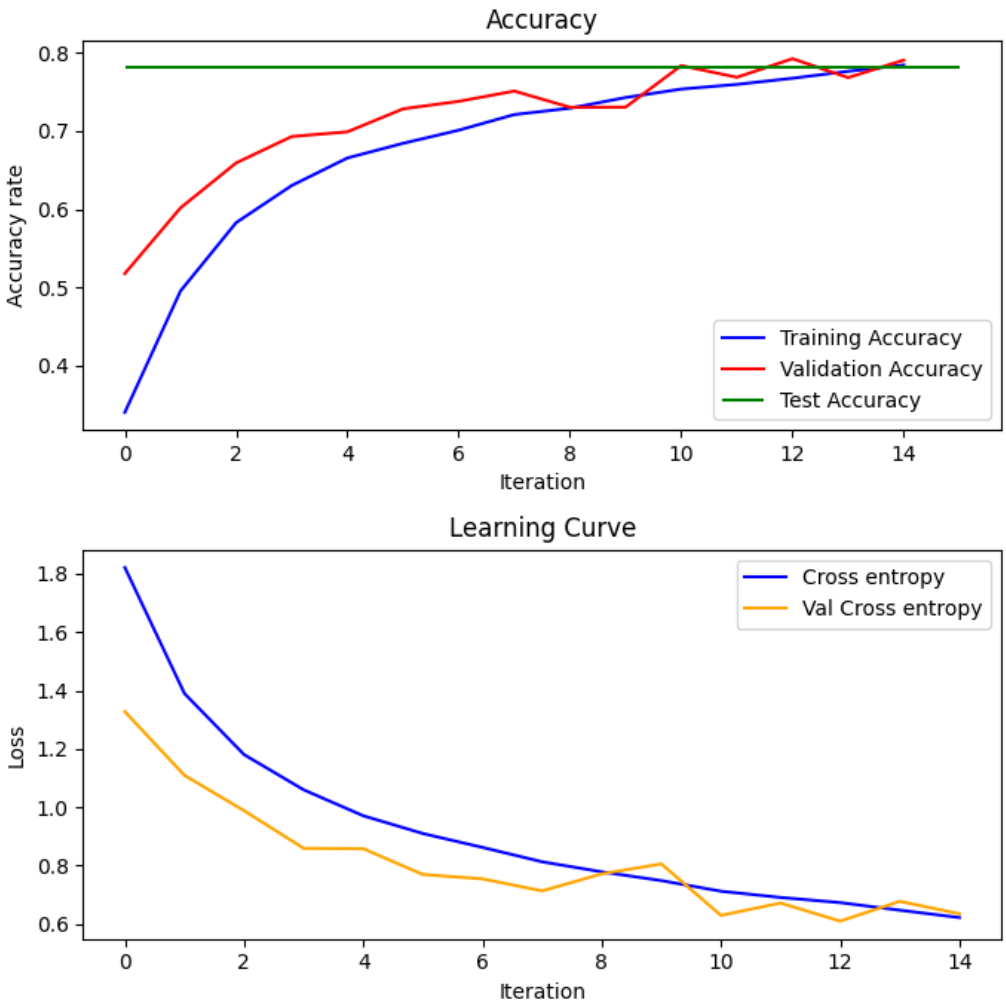

Figure 1: Training and validation accuracy/loss for CIFAR-10 baseline model.

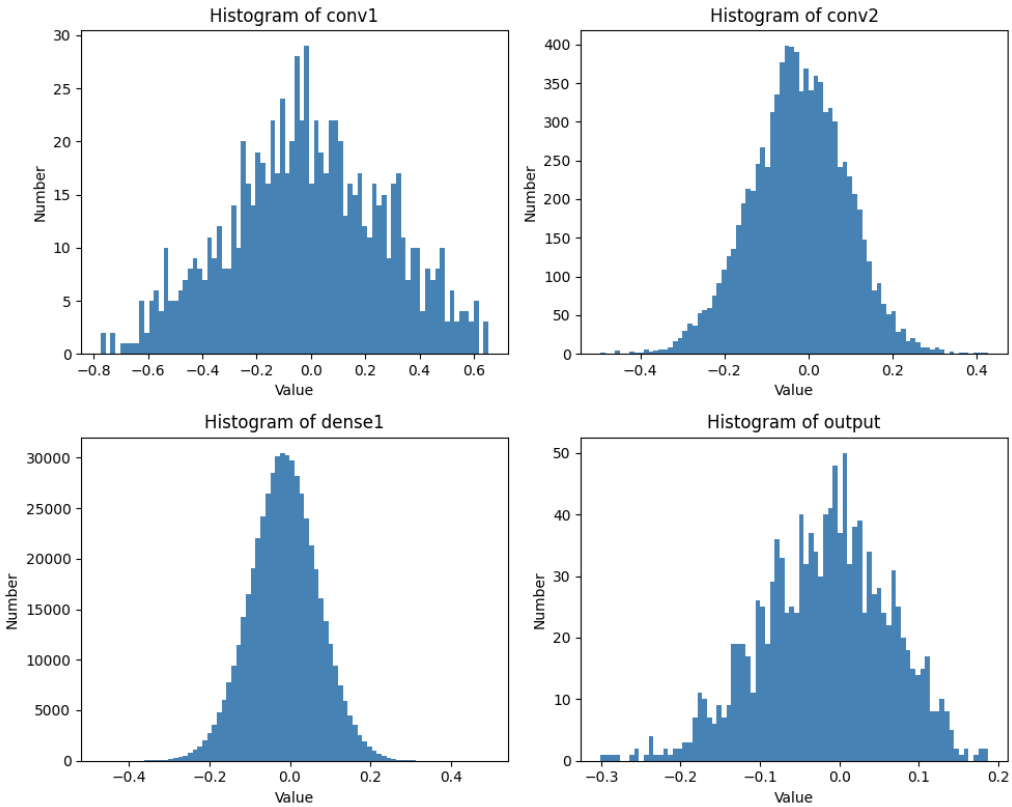

Figure 2: Distribution of weights - baseline model.

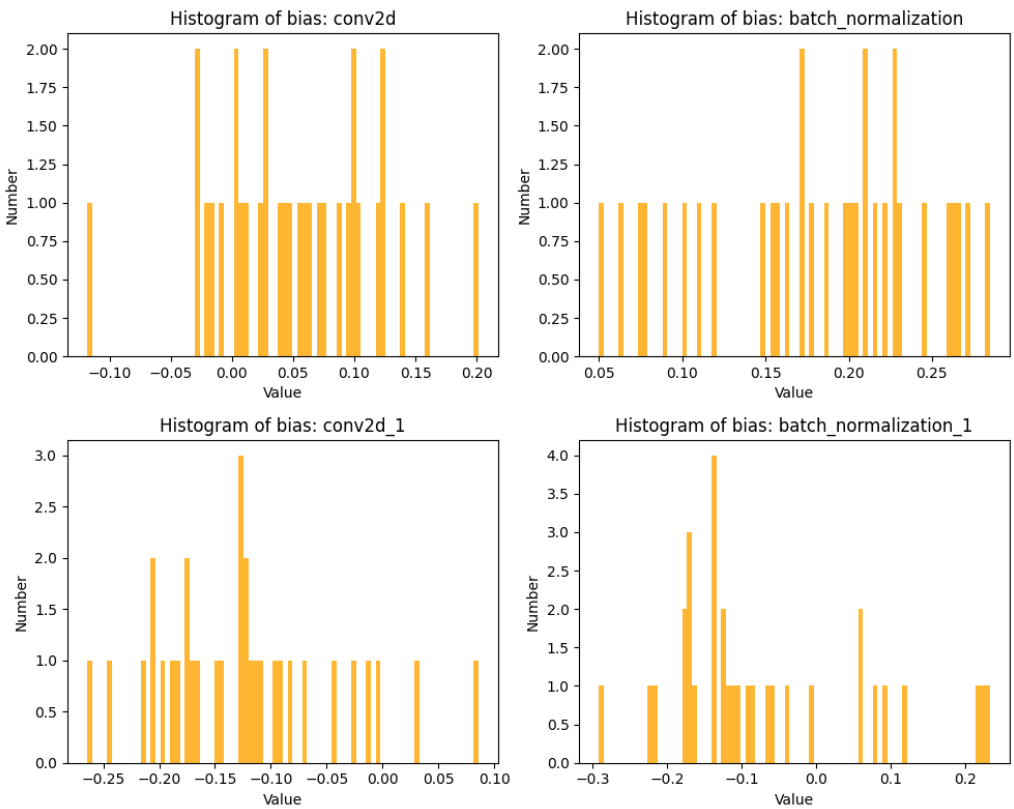

Figure 3: Distribution of biases - baseline model.

4.3 With (5,5) Filter and (2,2) Stride - Model Curves

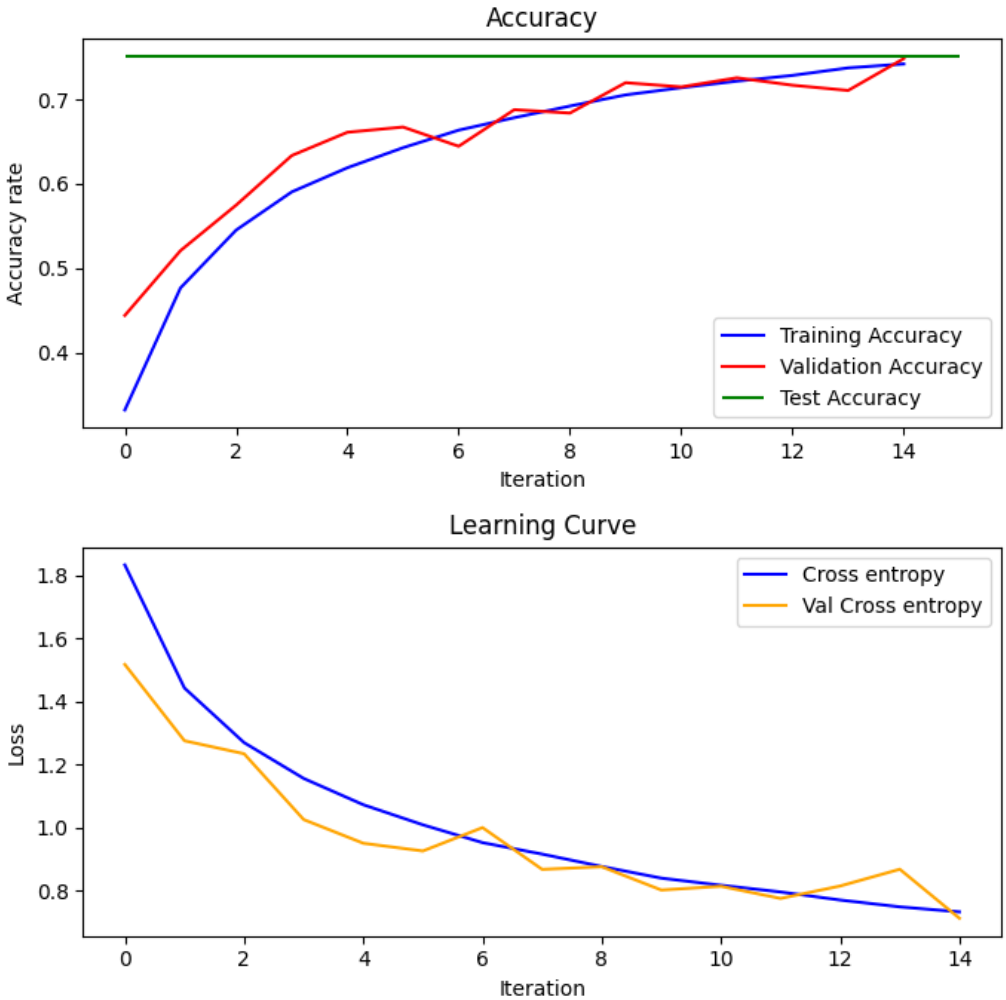

Figure 4: Training and validation accuracy/loss.

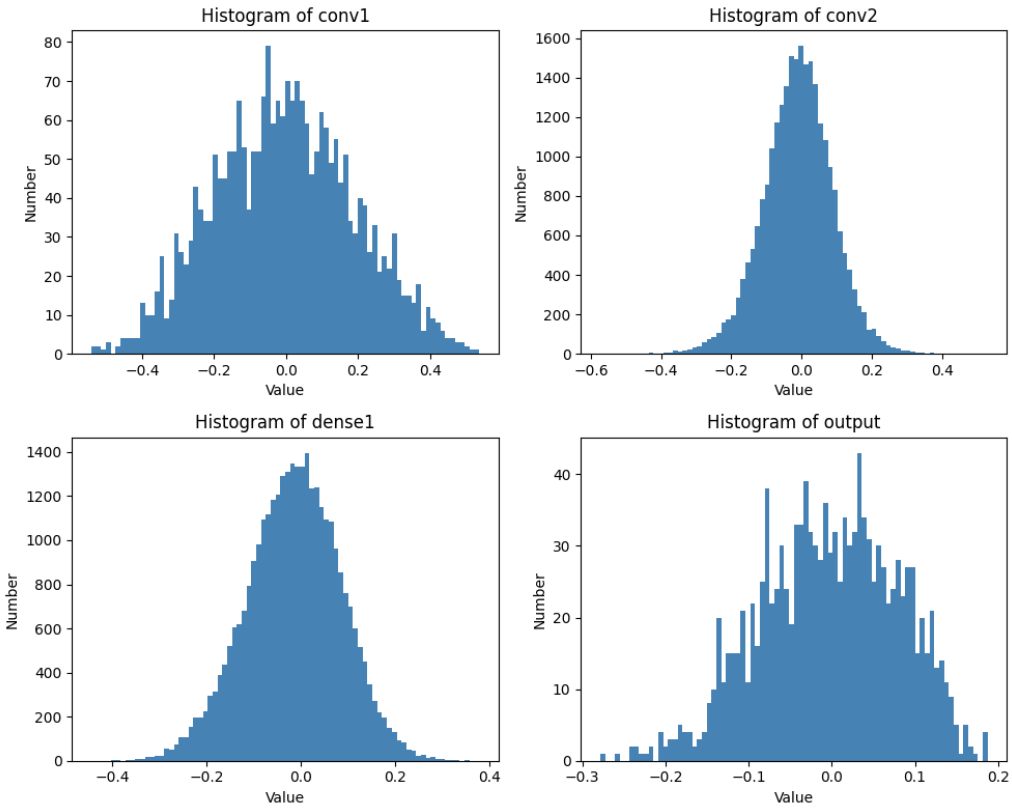

Figure 5: Distribution of weights .

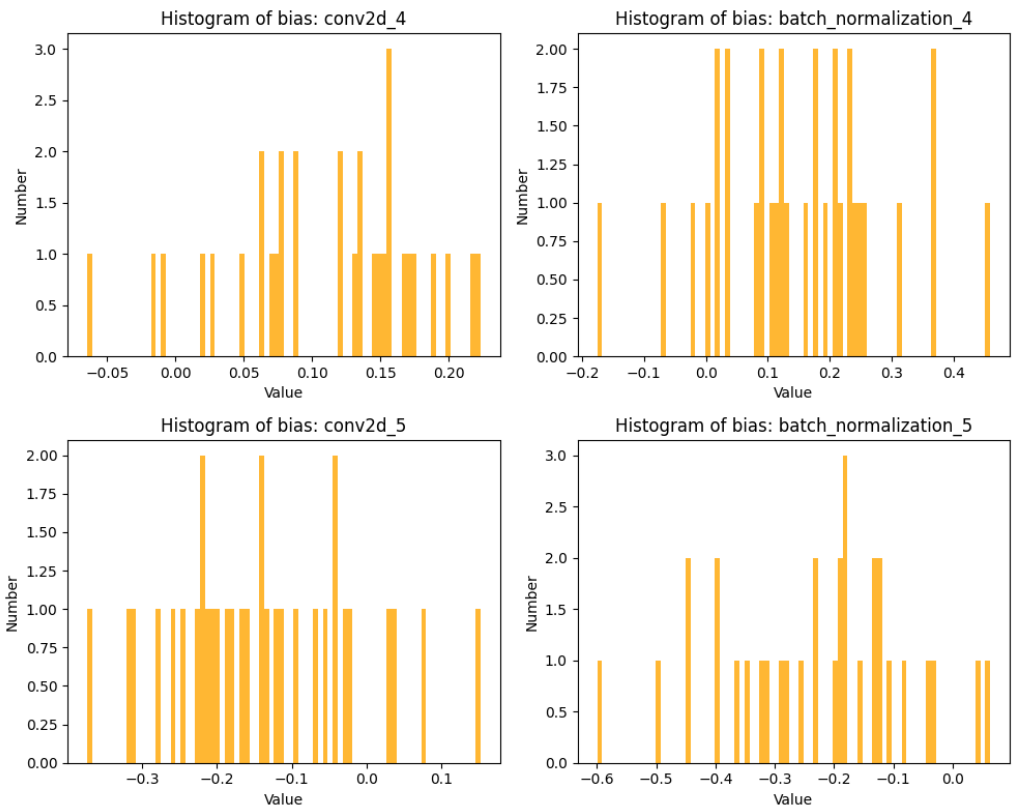

Figure 6: Distribution of biases - baseline.

4.4 With (7,7) Filter and (1,1) Stride - Model Curves

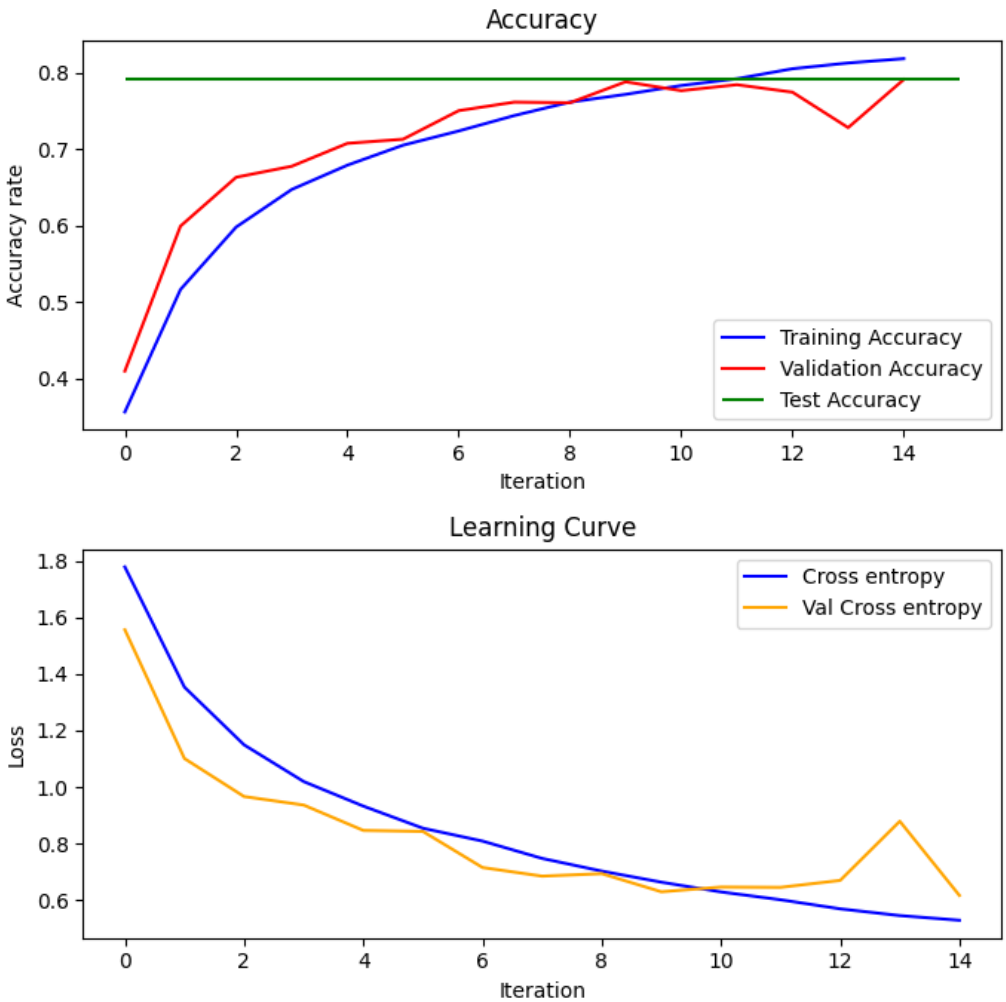

Figure 7: Training and validation accuracy/loss.

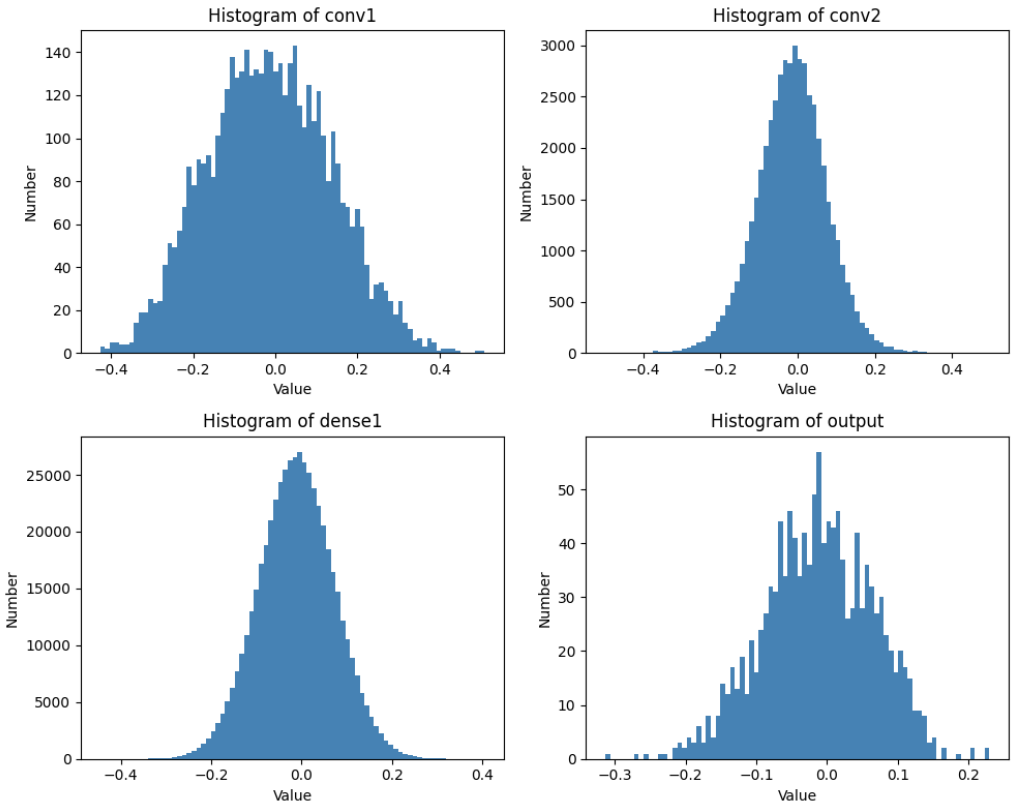

Figure 8: Distribution of weights.

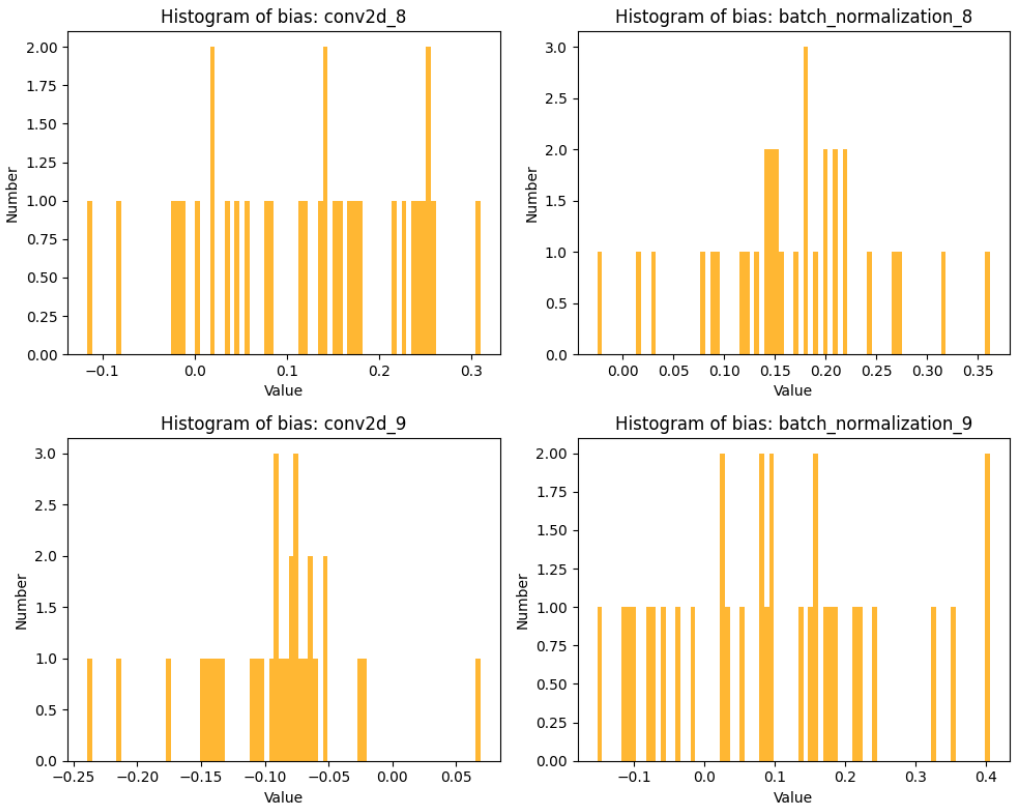

Figure 9: Distribution of biases .

4.5 Correctly and Misclassified Examples

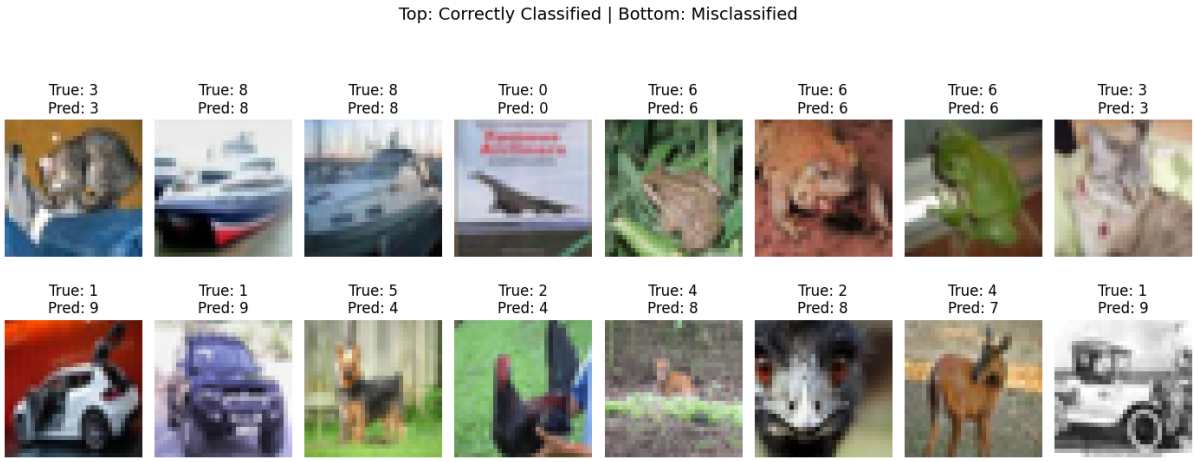

Figure 10: Correctly classified (top) and misclassified (bottom) examples for L2 model.

Most errors occur between visually similar classes (e.g., cat vs. dog, automobile vs. truck). The model captures coarse object shapes but occasionally misinterprets color or texture cues.

4.6 Feature Map Visualization

Feature maps show progressively abstracted representations:

- **Early layers:** capture local color gradients and edges.
- **Middle layers:** combine low-level features into patterns and textures.
- **Deep layers:** highlight object parts, focusing attention on discriminative regions.

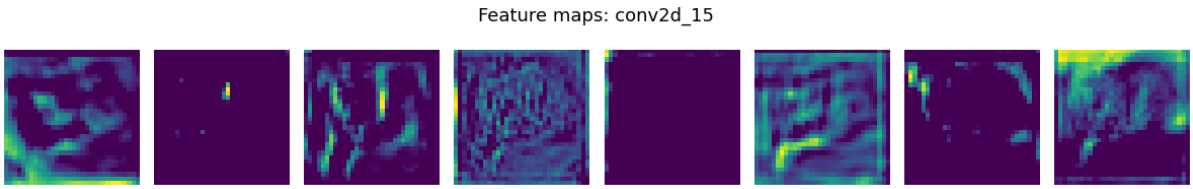

Figure 11: Sample feature maps from various convolutional layers of the L2 model.

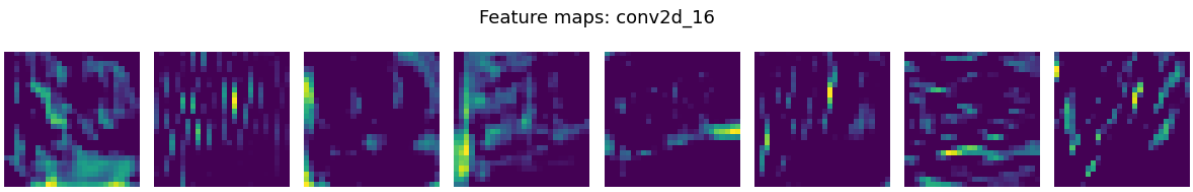

Figure 12: Sample feature maps from various convolutional layers of the L2 model.

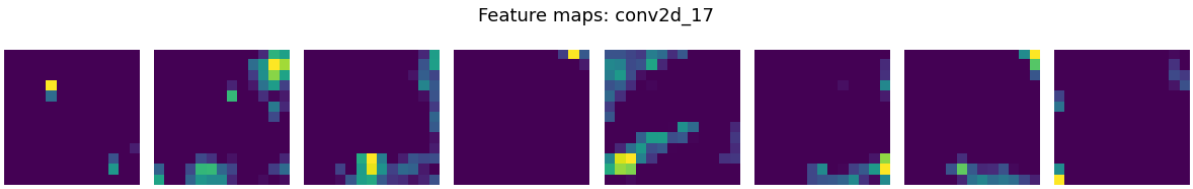

Figure 13: Sample feature maps from various convolutional layers of the L2 model.

4.7 Impact of L2 Regularization

L2 regularization reduced the magnitude of weights and activations, leading to smoother and more uniform feature maps. While training accuracy decreased, validation loss was more stable, indicating better generalization and less overfitting.

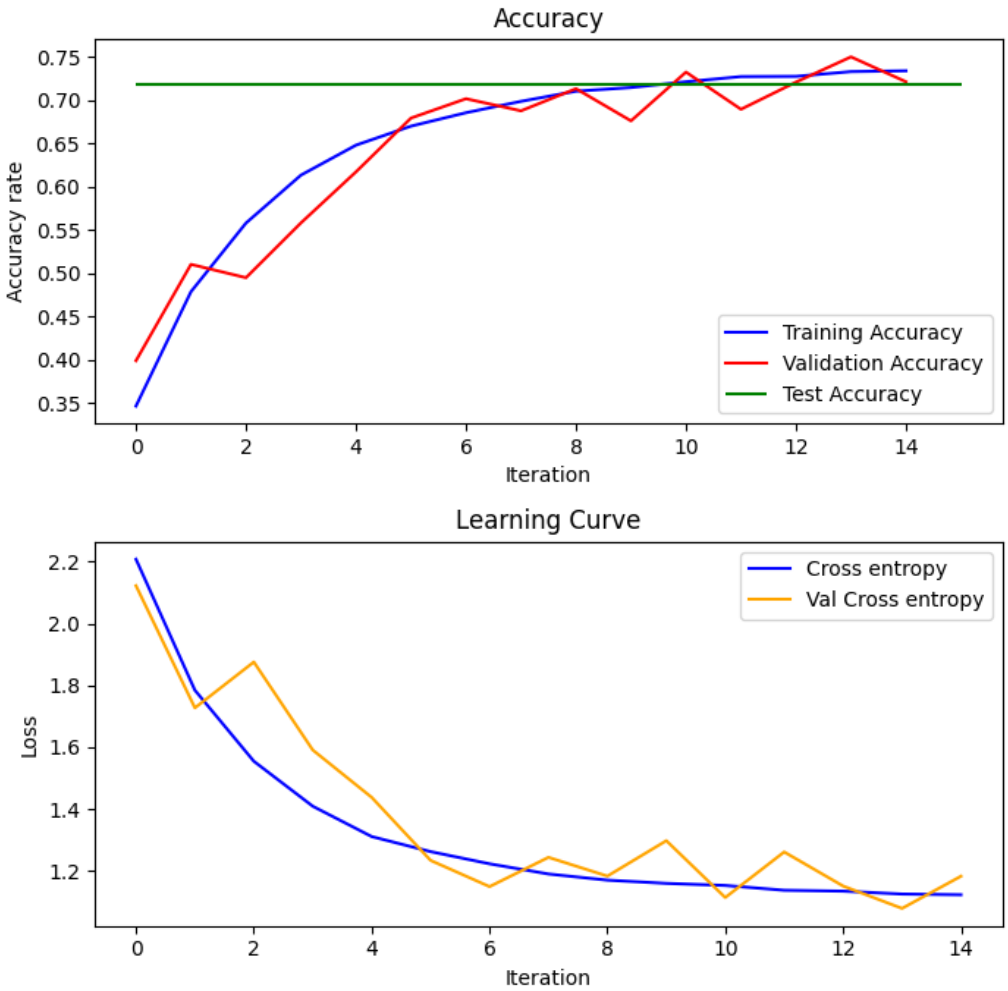

Figure 14: Training and validation accuracy/loss.

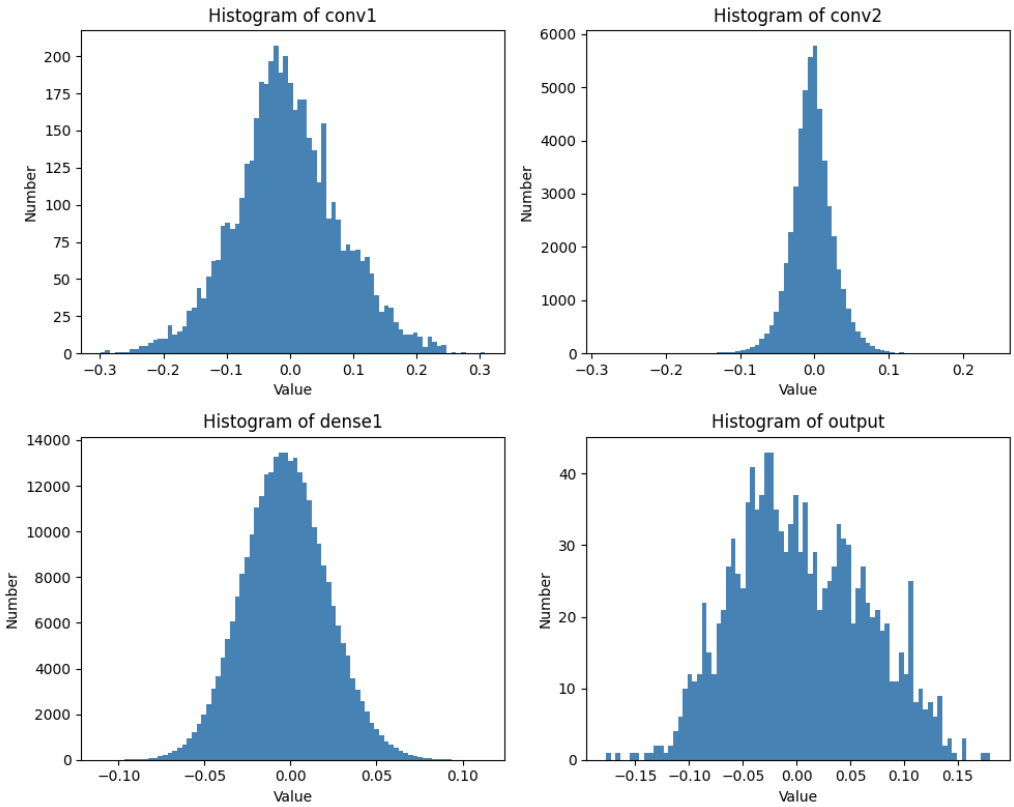

Figure 15: Distribution of weights.

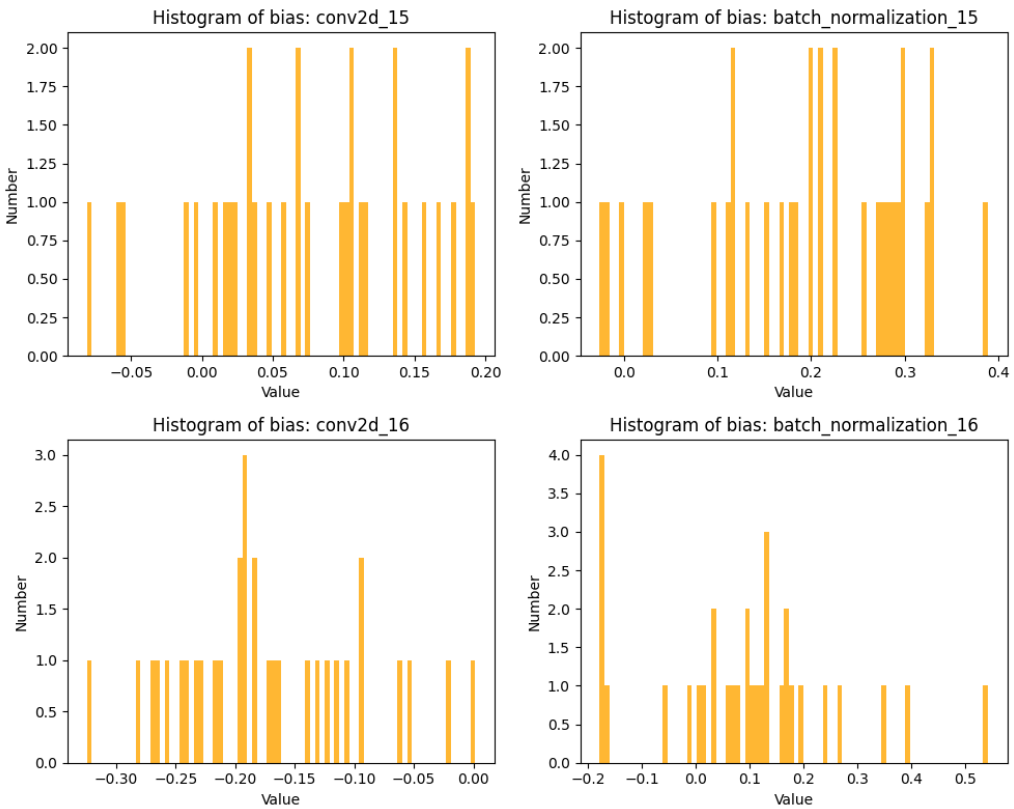

Figure 16: Distribution of biases .

5 Conclusion

The CNN achieved around 79% validation accuracy on CIFAR-10 using a simple two-block architecture. Filter size and stride significantly affect performance: small kernels with stride 1 perform best on small images. L2 regularization introduces stability and smoother activations but may require tuning to balance bias and variance. Data augmentation and normalization remain crucial for generalization in limited-data settings. Overall, (3×3, stride 1) with BatchNorm, Dropout, mild augmentation, and moderate L2 yields the best trade-off between accuracy, speed, and robustness.